

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 25

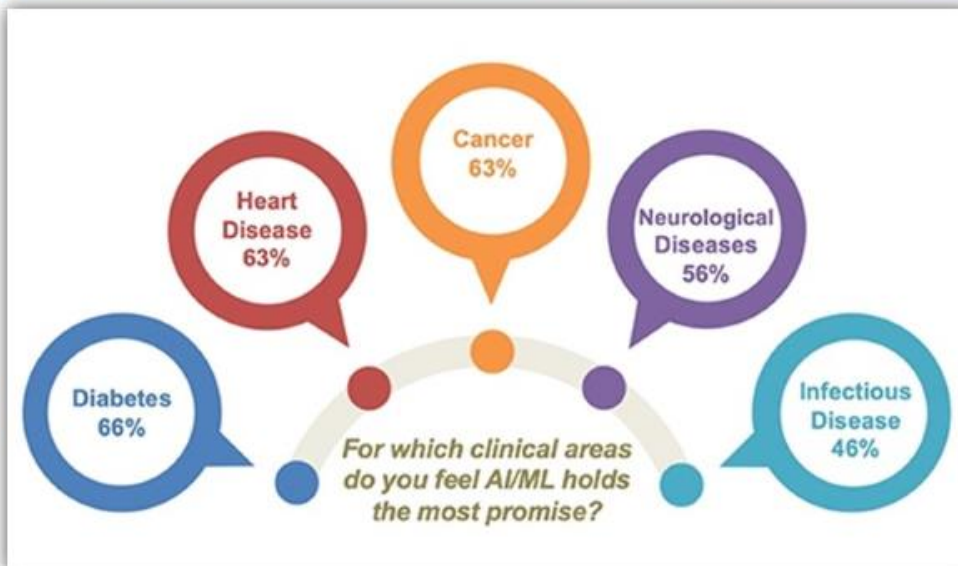
INFORMATION & COMMUNICATION TECHNOLOGY (ICT) - 6

APPLICATIONS OF AI - AGRICULTURE

- AI holds the promise of driving a food revolution and meeting the increased demand for food. We need to produce 50% more food to cater to an additional 2 billion people by 2050.
- AI has the potential to address challenges such as demand prediction, controlling the overuse / misuse of pesticides and fertilizers, improvement in crop yield through real time monitoring of crop health and advisories to farmers, advanced detection of pest attacks, monitoring of weather patterns and accurate weather prediction and prediction of crop prices to inform sowing practices.



- Application of AI in healthcare can help address issues of high barriers to access to healthcare facilities, particularly in rural areas that suffer from poor connectivity and limited supply of healthcare professionals.
- This can be achieved through implementation of use cases such as AI driven diagnostics, personalized treatment, early identification of potential pandemics, and imaging diagnostics, among others.



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APPLICATIONS OF AI - EDUCATION

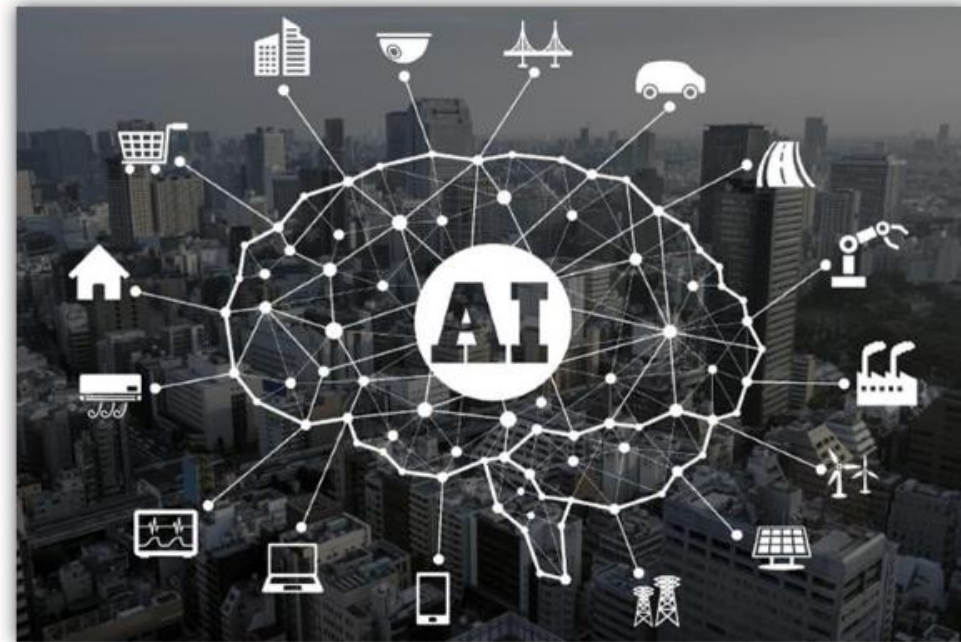
Applications of AI in education include augmenting and enhancing the learning experience through personalised learning, automated and smart test evaluation of tests, and predicting the need for student intervention to reduce dropouts.



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OTHER APPLICATIONS OF AI

- The other fields which can benefit immensely from AI include Automobiles
 - ✓ Automobiles
 - ✓ Finance and Economics
 - ✓ Cyber security
 - ✓ Government and Judicial Administration
 - ✓ Military
 - ✓ Hospitality



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GLOBAL DEVELOPMENTS IN AI

- Countries around the world are becoming increasingly aware of the potential economic and social benefits of developing and applying AI. China and U.K. estimate that 26% and 10% of their GDPs respectively in 2030 will be sourced from AI-related activities and businesses.
- There has been tremendous activity concerning AI policy positions and the development of an AI ecosystem in different countries over the last 3-4 years.
- Infrastructural supply side interventions have been planned by various countries for creating a larger ecosystem for AI development.
- Many countries have instituted dedicated public offices such as Ministry of AI (UAE) and AI Council (U.K.), while China and Japan have allowed existing ministries to take up development and implementation of AI.



**HIS EXCELLENCY
OMAR SULTAN AL OLAMA**
Minister of State for Artificial Intelligence, UAE

- The Global Partnership on Artificial Intelligence (GPAI) is a multi-stakeholder initiative. It aims to use the technology to bridge the gap between theory and practice on AI through research and applied activities on AI-related priorities.
- GPAI is the implementation of the idea developed within the G7. It was launched in June, 2020.
- The founding members of GPAI are:

✓ Australia	✓ European Union	✓ Singapore
✓ Canada	✓ United Kingdom	✓ Slovenia
✓ France	✓ United States	✓ Italy
✓ Germany	✓ New Zealand	✓ Japan
✓ India	✓ Republic of Korea	✓ Mexico



Brazil, Netherlands, Poland and Spain joined GPAI in December 2020.

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SIGNIFICANCE OF GPAI

- GPAI is helpful in providing a platform for sharing research and identifying key issues among AI practitioners.
- It works for facilitating an international collaboration which reduces duplication and acts as a global reference point for specific AI issues as well. It ultimately leads to adoption of trustworthy AI.
- GPAI assesses scientific, socio economic information collected by it on a comprehensive, objective, open, and transparent basis.
- This information is relevant for understanding AI impacts, leading to its development and options for adaptation and mitigation of potential challenges.

**GLOBAL PARTNERSHIP
ON ARTIFICIAL
INTELLIGENCE
(GPAI)**



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